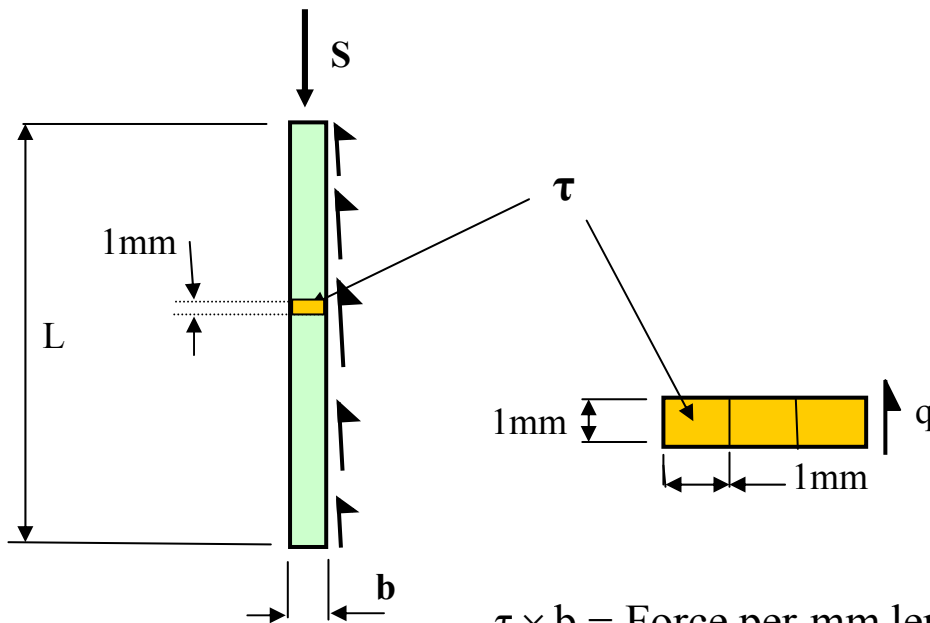


BOX BEAM SHEAR FLOWS.....SHEAR FLOW (q) BASICS

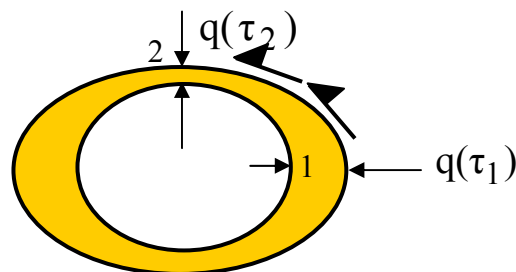


$$\tau \times b = \text{Force per mm length} = q \text{ N/mm}$$

$$\text{say } b = 3\text{mm then } q = 3 \times \tau$$

$$\int_0^L q \times dx = S$$

For a section subjected to torque, the shear flow q is constant around the section hence the shear stress must vary with wall thickness such that the stress thickness product remains constant.

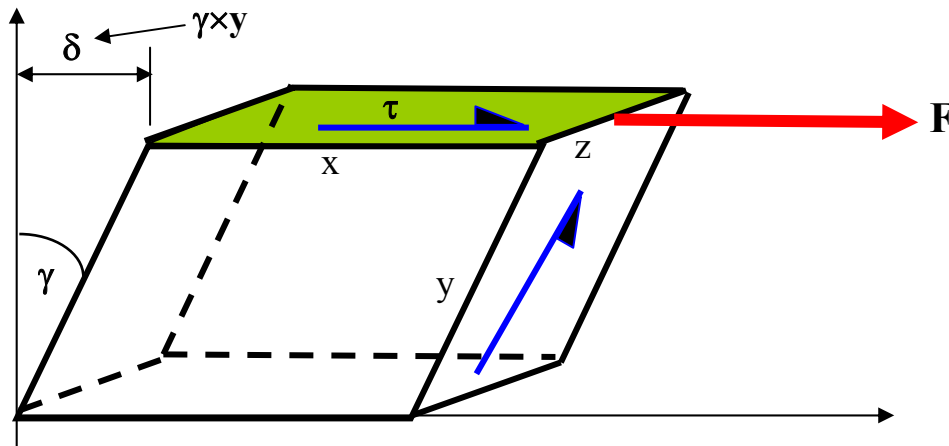


$$\tau_1 \leq \tau_2$$

TWIST OF A THIN WALLED TUBE

Strain energy in shear (U)

Consider a block of material under a shear force F and a shear stress τ

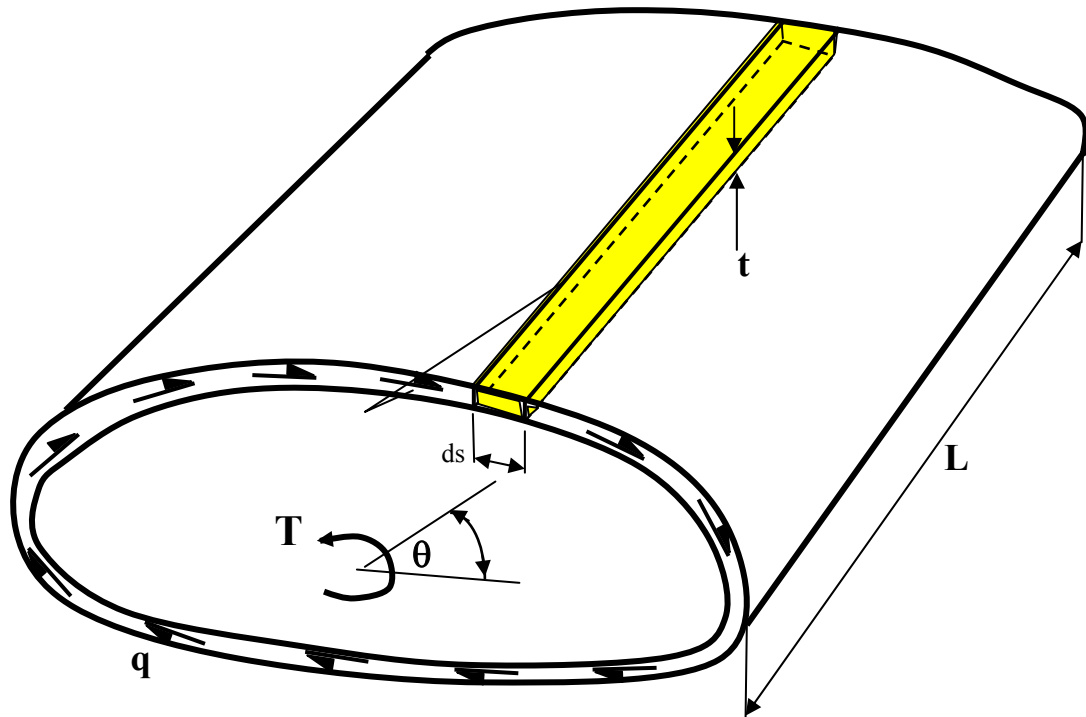


$$\tau = \frac{F}{xz}$$

work done = strain energy in block (U)

$$\begin{aligned} \frac{F}{2} \times y \times \gamma &= \frac{1}{2} \times (F \times y) \times \gamma \\ &= \frac{1}{2} \times \text{couple} \times \text{angle turned through} \\ &= \frac{1}{2} \times (\tau xz y) \gamma \quad \text{but} \quad \gamma = \frac{\tau}{G} \end{aligned}$$

$$\text{Therefore } U = \frac{\tau^2}{2G} \times \text{VOLUME}$$



The strain energy of length L of the tube is:-

$$U = \int_0^s \left(\frac{\tau^2}{2G} \right) \underbrace{L t ds}_{\text{VOLUME}} = \int_0^s \frac{q^2 L}{2Gt} ds \quad \text{since } \tau = \frac{q}{t}$$

$U = \text{work done by torque}$

$$= \frac{1}{2} T \theta \quad \text{where } \theta = \text{angle of twist in radians}$$

$$= \frac{1}{2} (2Aq) \theta \quad \text{since } T = 2Aq$$

$$\text{therefore } Aq\theta = \int_0^s \frac{q^2 L}{2Gt} ds \quad \text{or}$$

$$\theta = \frac{L}{2AG} \int_0^s \frac{q}{t} ds$$

BOX BEAM SHEAR FLOWS

Shear Flow in Thin Curved Webs

Shear flow in plain webs has already been discussed in the section on shear stress in beams. In this section we are now concerned with curved webs, such as found in aircraft wings, fuselage and stressed skin construction in general.

Here we are interested in determining magnitude, direction and location of the shear force.

a) Magnitude and direction of shear force

Consider a thin curved web under a shear stress τ (N/mm²) let q be the shear flow in N/mm where $\tau = q/t$; t = thickness of the web

Reference figure 1, consider an elemental length dL . The force on this element is $q \cdot dL$ and the components of this force are;

$q \cdot dx$ horizontally

$q \cdot dy$ vertically the

total horizontal force = $\int_0^x q \cdot dx = qX$ and

total vertical force = $\int_0^y q \cdot dy = qY$

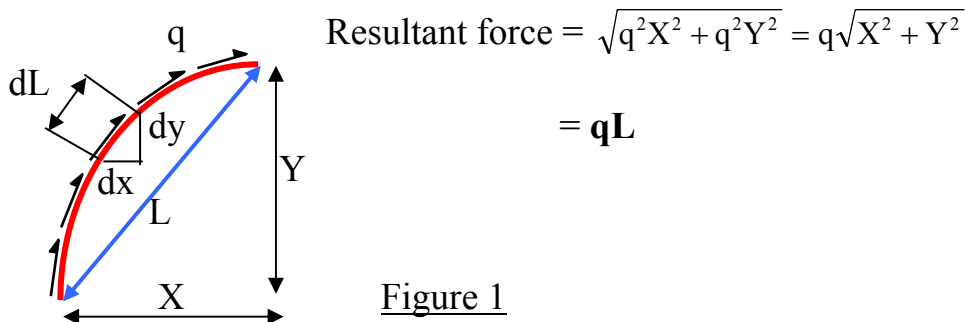
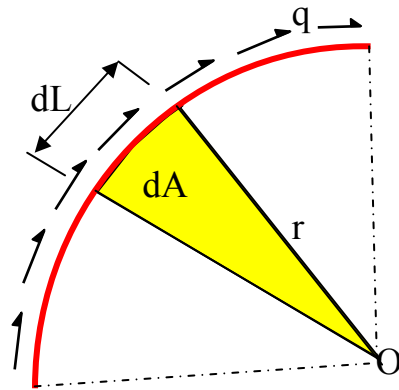


Figure 1

The dimensions X and Y are the projected lengths of the curved web when taken over a straight line and in this case perpendicular to each other, although projections in any direction are valid so long as the lines from each end of the web are parallel. If a line is drawn joining the ends of the web, L , for this there will be no force perpendicular to the line, consequently the component parallel to the line will be the resultant shear force, this conforms to the polygon of forces where the line joining the ends of the web is the closing force requires to complete the polygon.

b) Moment of shear force at any point

Consider the moment of the shear flow about a point O.



Turning moment of a force $q.dL = q.dL.r$
where r = the moment arm

$$\text{but } dA = (1/2)dL.r \text{ or } dL.r = 2.dA$$

$$\therefore \text{Total turning moment } T = \sum q.r.dL \\ = \sum 2q.dA$$

$$\text{i.e. } \underline{T = 2Aq}$$

where A is the total area enclosed by the web and lines from the end of the web to O.

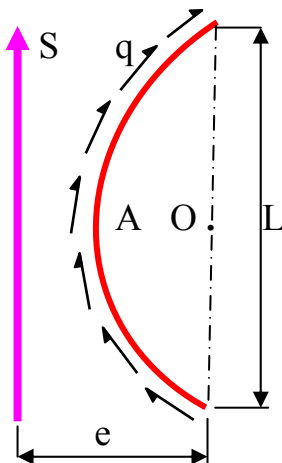
Figure 2

$$\underline{\text{Therefore } T = 2At \tau}$$

Note, the shear stress in the web can be determined from the formula $\tau = T/2At$, this is called the Bredt-Batho formula.

c) Location of resultant shear force

From the above we have determined both the magnitude and direction of the resultant shear force, and established that there is a definite moment about the point O. Taking O to lie on line L (see figure 3), then joining the ends of the curved beam it is seen that as a moment exists at O (equal to $2Aq$) that the resultant shear force cannot act along the line L but must have a line of action some distance e from O as shown in the diagram.



The distance e can be determined by taking moments about O such that;

$$S.e = 2Aq$$

$$e = 2.Aq/S$$

we have shown $S = qL$

$$\text{therefore } \underline{e = 2A/L}$$

Figure 3

e gives the position of the shear centre. If the load is applied through the shear centre then no twisting of the section will occur. Note that the shear centre is a geometric property of the section.

Example 1

(a) An extruded aluminium tube has the cross-section shown in figure 4

- (i) If the shear stress may not exceed 70 N/mm^2 , what is the maximum permissible torque based on a safety factor of 2.5 which can be applied.

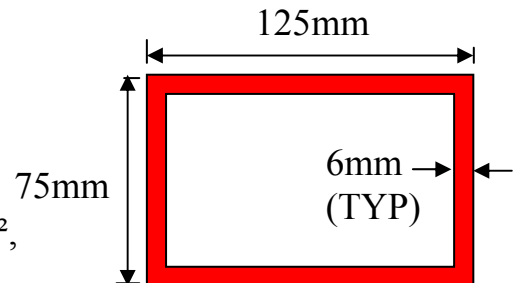


Figure 4

- (ii) Determine the angle of twist per metre length of the tube.
 $G = 27 \text{ kN/mm}^2$.

Solution

- (a) maximum working stress $= 70/2.5 = 28 \text{ N/mm}^2$
mean area $= (125 - 6) \times (75 - 6) = 8211 \text{ mm}^2$
shear flow $= \text{shear stress} \times \text{width} = 28 \times 6 = 168 \text{ N/mm}$
therefore Torque $= 2 \times 8211 \times 168 = 2,758,896 \text{ Nmm} = \underline{\underline{2.76 \text{ kNm}}}$

- (b) For this part the strain energy equation applies namely:-

$$\theta = \frac{L}{2AG} \sum \frac{q\Delta L}{t} \text{ (radians)} = \frac{L}{2AG} \left(\frac{q_1 L_1}{t_1} + \frac{q_2 L_2}{t_2} + \frac{q_3 L_3}{t_3} + \frac{q_4 L_4}{t_4} \right)$$

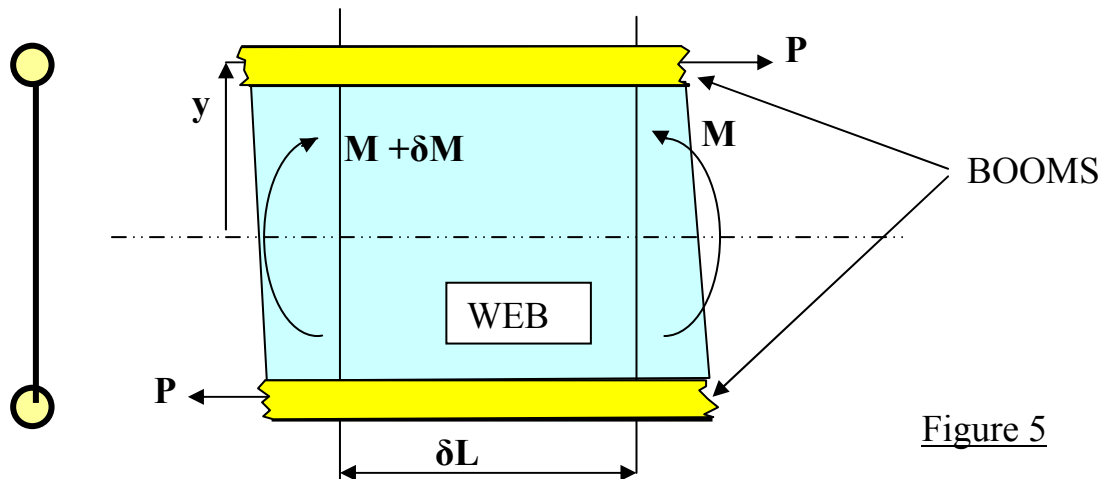
$$= \frac{1000}{2 \times 8211 \times 27 \times 10^3} \left(\frac{168 \times 69}{6} + \frac{168 \times 119}{6} + \frac{168 \times 69}{6} + \frac{168 \times 119}{6} \right)$$

$\theta = 0.024 \text{ radians} = 1.36 \text{ degrees}$

See Tutorial Question 1

END LOADS IN BOOMS

Consider a section of a simple boom/web beam as shown in Figure 5.



Assuming that the idealised booms resist the bending stress and that the skin carries the shear stresses then the change in end load in the booms is given by :-

$$\delta P = \frac{(M + \delta M) \times Ay}{I} - \frac{M \times Ay}{I} \quad \text{from} \quad \sigma = \frac{P}{A} = \frac{My}{I}$$

$$\delta P = \frac{\delta M \times Ay}{I} \quad \text{where } A = \text{c.s.a. of the boom}$$

I = total second moment of area of the beam

Now the change of end load per unit length of the beam

$$= \delta P / \delta L = \frac{\delta M \times Ay}{\delta L \times I}$$

when $\delta L \rightarrow 0$ $\delta M / \delta L \rightarrow dM/dL = \text{the shear force } F$

then

$$\Delta \mathbf{P} / \Delta \mathbf{L} = \mathbf{F}(\mathbf{A} \bar{\mathbf{y}}) / \mathbf{I}$$

This relationship gives the change in boom end load per unit length which can be used to determine the individual skin shear flows.

OPEN SECTIONED BEAMS

Example 2

Figure 6 shows a beam with a curved open section loaded through its shear centre with 50kN. Determine the shear flows in the three panel skins.

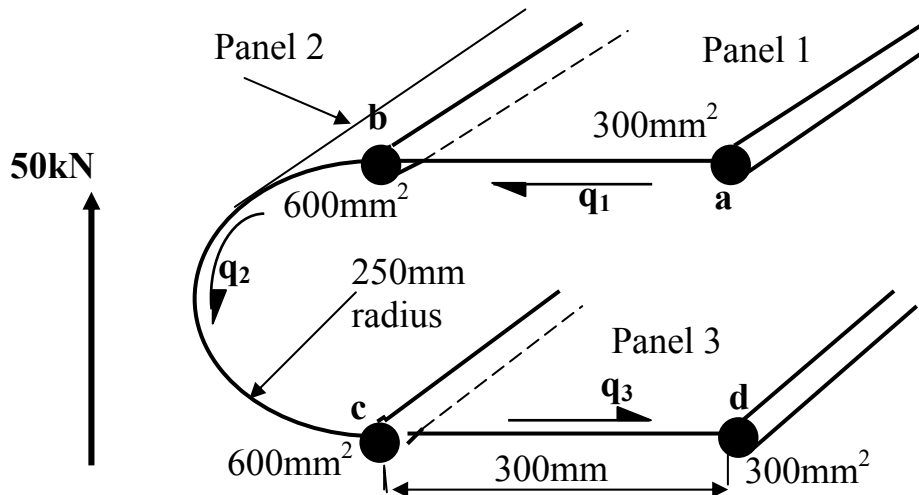


Figure 6

Using the equation $\Delta P/\Delta L = F(A\bar{y})/I$

where $I = \Sigma Ah^2 = 2(300 \times 250^2) + 2(600 \times 250^2) = 112.5 \times 10^6 \text{ mm}^4$

For booms a & d

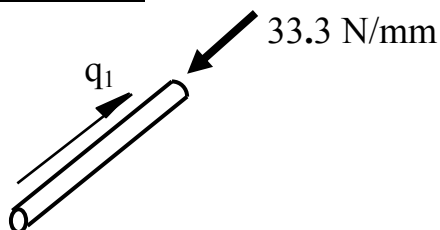
$$\Delta P/\Delta L = \frac{50 \times 10^3 (300 \times 250)}{112.5 \times 10^6} = 33.3 \text{ N/mm}$$

For booms b & c

$$\Delta P/\Delta L = 2 \times 33.3 = 66.6 \text{ N/mm (twice area of a & d)}$$

Looking at the spanwise equilibrium of each boom then:-

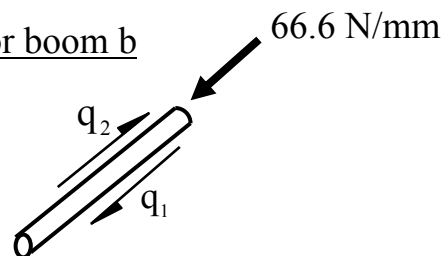
For boom a



$$33.3 - q_1 = 0$$

$$\text{hence } q_1 = \underline{\underline{33.3 \text{ N/mm}}}$$

For boom b



$$q_1 + 66.6 - q_2 = 0$$

$$q_2 = 33.3 + 66.6 = \underline{\underline{99.9 \text{ N/mm}}}$$

Negative values of the shear flows would mean that the flows are in the opposite direction to those assumed.

Boom c is similar to b & d is similar to a.

These skin shear flows (N/mm) are shown Figure 7.

The skin shear stress can be found from:- $\tau = q/t$ shear flow/skin thickness .

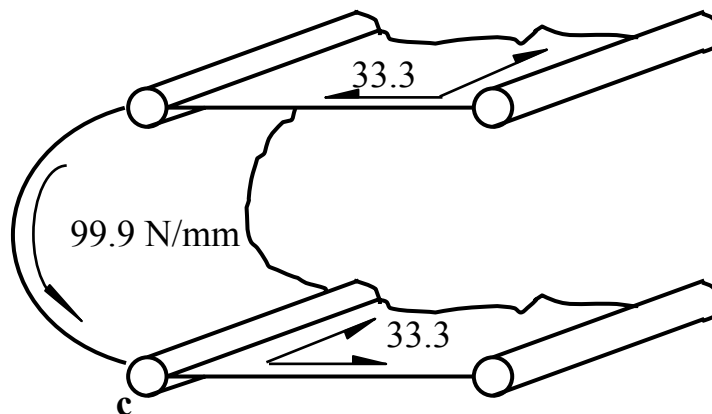


Figure 7

By taking moments about point (c) the shear centre of the beam section can be determined. The torque due to the load is balanced by the torque due to the shear flow.

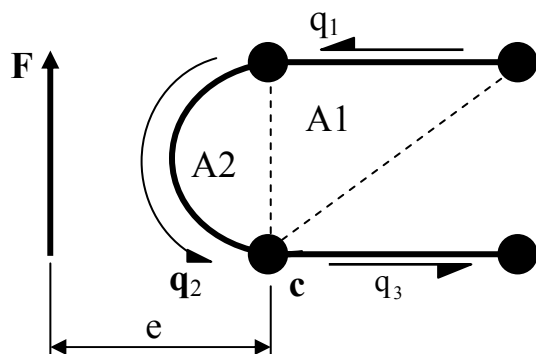


Figure 8

$$F \times e = \Sigma 2Aq \text{ (see figure 8)}$$

Moment due to q_1 and q_2 :-

$$(q_1) 2 \times 300 \times \frac{500}{2} \times 33.3 = 4995000$$

$$(q_2) 2 \times \frac{\pi}{2} \times 250^2 \times 100 = 19634954$$

$$\text{Total} = 24,629,954 \text{ Nmm}$$

$$\text{Therefore } e = \frac{24629954}{50 \times 10^3} = \underline{\underline{492.6\text{mm}}}$$

See Tutorial Question 2

CLOSED BOX SECTION BEAMS

The open sectioned beam previously considered is capable of resisting loads passing through the shear centre, but it is not suitable for resisting torsion. In many structures the loading is not through the shear centre and twisting will result. A closed box structure capable of resisting torsion is consequently used.

In such a box beam the torsion will induce an additional set of shear flows to those due to the shear force (which is causing bending). The method of analysis is to express the shear flow on the various skin elements in terms of one unknown shear flow " q_0 " by taking spanwise equilibrium for each boom/skin element, the unknown shear flow " q_0 " can then be found by moment equilibrium and then fed back to find the other unknown shear flows.

Example 3

Determine the shear flows in the webs and skins of the idealised box section shown in figure 9. $E = 65\text{kN/mm}^2$, $G = 25\text{kN/mm}^2$

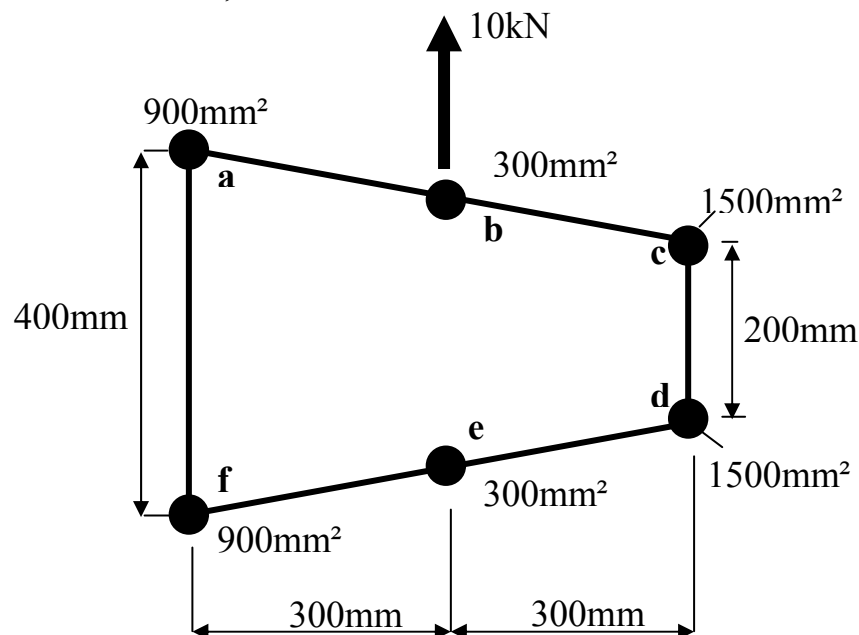


Figure 9

The applied change of end load for each boom can be found from the relationship:-
 $\Delta P / \Delta L = F(A\bar{y}) / I$

$$I = \sum Ah^2 = 2((900 \times 200^2) + (300 \times 150^2) + (1500 \times 100^2)) = 115.5 \times 10^6 \text{ mm}^4$$

Booms a & f

$$\Delta P / \Delta L = \frac{10 \times 10^3 \times (900 \times 200)}{115.5 \times 10^6} = 15.6 \text{ N/mm}$$

Booms b & e

$$\Delta P / \Delta L = \frac{10 \times 10^3 \times (300 \times 150)}{115.5 \times 10^6} = 3.9 \text{ N/mm}$$

Booms c & d

$$\Delta P / \Delta L = \frac{10 \times 10^3 \times (1500 \times 100)}{115.5 \times 10^6} = 13.0 \text{ N/mm}$$

Consider the shear flow to be clockwise around the face of the box as shown in Figure 10.

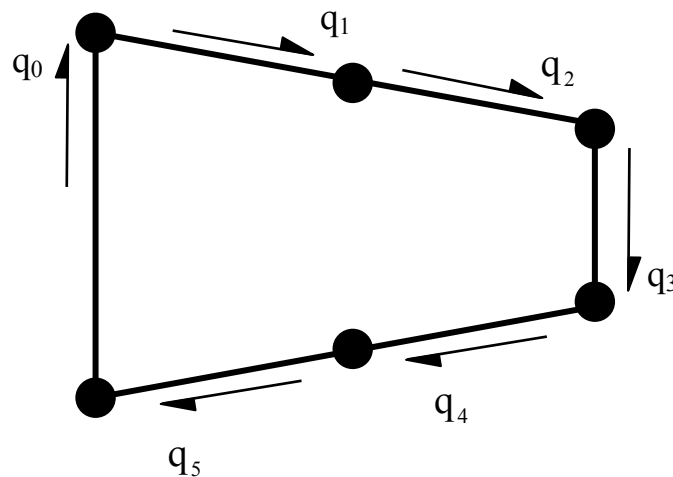
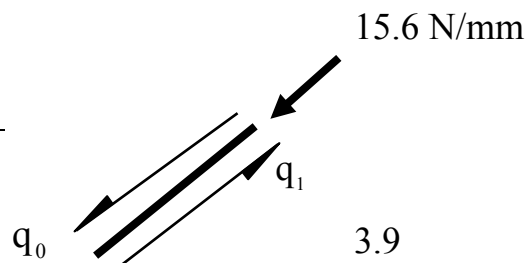


Figure 10

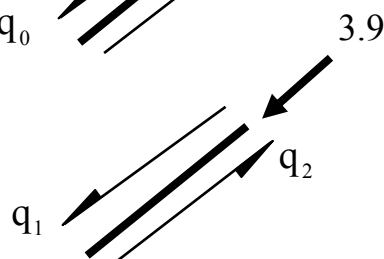
Complementary shear flows in the spanwise direction can now be drawn:-

Booms a & f



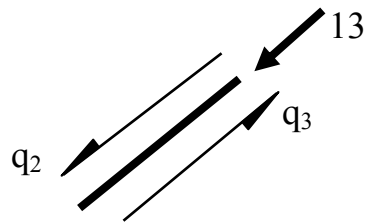
hence $q_1 = q_0 + 15.6$ (equ.1)

Booms b & e



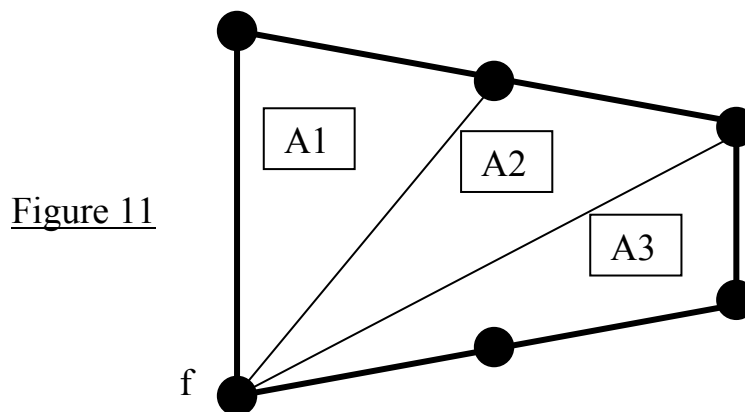
hence $q_2 = q_1 + 3.9$
 $q_2 = q_0 + 15.6 + 3.9$
 $q_2 = q_0 + 19.5$ (equ.2)

Booms c & d



hence $q_3 = q_2 + 13$
 $= q_0 + 32.5$ (equ.3)

To determine q_0 , moments must be taken about some point in the section.
 For moments about point f (see Figure 11)



$A1 = A2 = A3 = 6 \times 10^4 \text{ mm}^2$ (areas in this case are equal by co-incidence)

$$\begin{aligned} 10000 \times 300 &= \Sigma 2Aq \\ &= 2 \times 6 \times 10^4 (q_1 + q_2 + q_3) \\ &= 2 \times 6 \times 10^4 (q_0 + 15.6 + q_0 + 19.5 + q_0 + 32.5) \end{aligned}$$

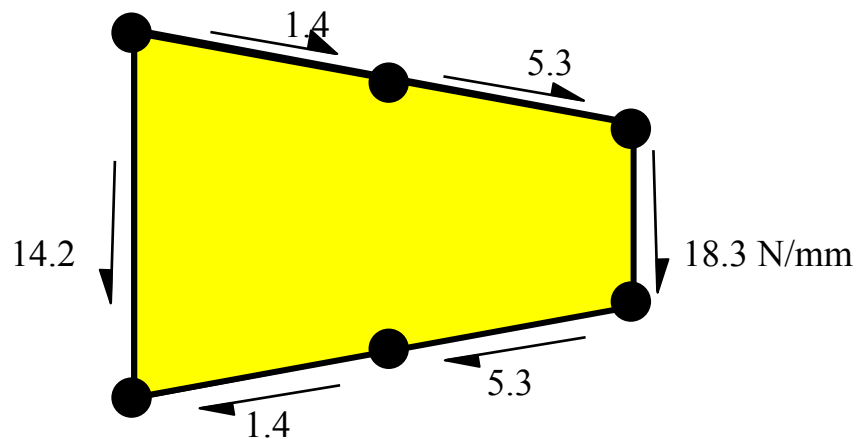
(divide each side by 12) $25 = 3q_0 + 67.6$
 $q_0 = -14.2 \text{ N/mm}$

Feeding this value back into equations 1,2 and 3 we get:-

$$\begin{aligned} q_1 &= q_0 + 15.6 = -14.2 + 15.6 = \underline{1.4 \text{ N/mm}} \\ q_2 &= q_0 + 19.5 = -14.2 + 19.5 = \underline{5.3 \text{ N/mm}} \\ q_3 &= q_0 + 32.5 = -14.2 + 32.5 = \underline{18.3 \text{ N/mm}} \end{aligned}$$

These shear flows are shown in Figure 12

Figure 12



Evaluation of the twist of the beam

From
$$\theta = \frac{L}{2AG} \sum \frac{q\Delta L}{t}$$

ΔL = Length over which shear flow acts

Take skin thickness = 1mm, web thickness = 2mm

$$\theta = \frac{1000}{2 \times 18 \times 10^4 \times 25 \times 10^3} \left(\frac{q_0 L_0}{t_0} + \frac{q_1 L_1}{t_1} + \dots + \frac{q_n L_n}{t_n} \right)$$

$$\theta = 11 \times 10^{-8} \left(\frac{-14.2 \times 400}{2} + \frac{2 \times 1.4 \times 304}{1} + \frac{2 \times 5.3 \times 304}{1} + \frac{18.3 \times 200}{2} \right)$$

$\theta = 3.37 \times 10^{-4}$ (0.019 degrees)

See Tutorial Question 3

If it is required to evaluate the shear flows for a number of loading conditions then it is best to find the shear centre and determine the shear flows due to:-

- 1) Vertical force at the shear centre
- 2) Torque at the shear centre

Any number of loads can be transformed to the shear centre, the total shear flow being found by superposition. The direct force can be solved from

$$q = \frac{F A \bar{y}}{I} \quad \text{and the torque from} \quad q = \frac{T}{2A}$$

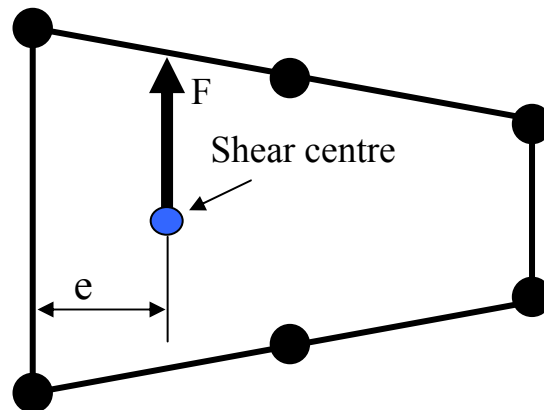


Figure 13

First the shear centre (e) shown in Figure 13 must be found :-

- Take a unit force F (say 1kN)
- Calculate I for idealised box
- Calculate shear flow for each boom
- Determine the shear flows in terms of q_0
- Calculate q_0 using $\theta = 0$
- Calculate location of shear centre by taking moments

So still using example 3, $I = 115.5 \times 10^6$ and for unit load:-

$$q_1 = q_0 + 1.56, \quad q_2 = q_0 + 1.95 \quad \text{and} \quad q_3 = q_0 + 3.25$$

$$\theta = 0 = \frac{L}{2AG} \sum \frac{q\Delta L}{t} = \frac{q_0 \times 400}{2} + \frac{2 \times 304 \times q_1}{1} + \frac{2 \times 304 \times q_2}{1} + \frac{q_3 \times 200}{2}$$

$$q_0 = -1.62 \text{ N/mm}$$

$$q_1 = -0.06 \text{ N/mm}$$

$$q_2 = 0.33 \text{ N/mm}$$

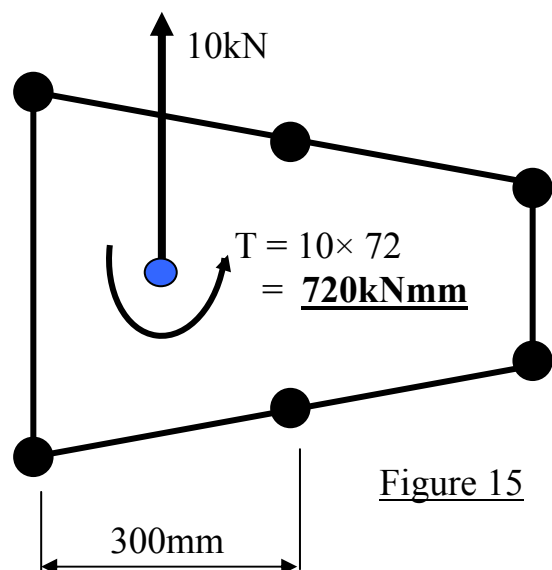
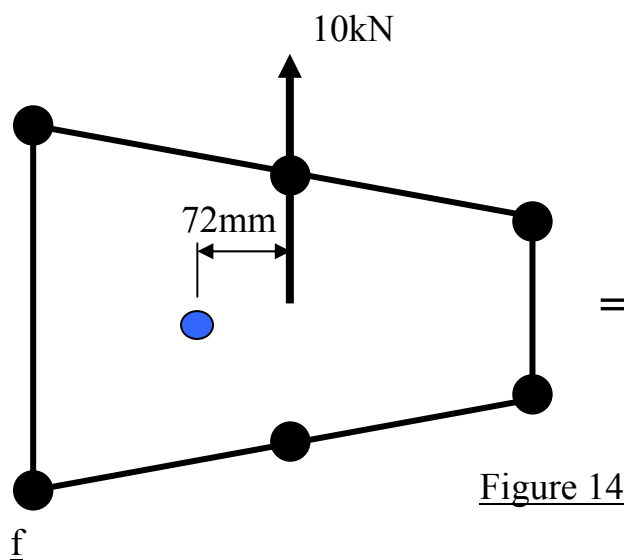
$$q_3 = 1.63 \text{ N/mm}$$

**Shear flows for a 1kN
load acting through the
shear centre**

Take moments about point f:-

$$\begin{aligned} 1000e &= \Sigma 2Aq \\ &= 2A_1q_1 + 2A_2q_2 + 2A_3q_3 \\ &= 2 \times 6 \times 10^4 (-0.06 + 0.33 + 1.63) \\ \mathbf{e} &= \mathbf{228\text{mm}} \end{aligned}$$

If 10 kN is applied at the centre of the box as shown in Figure 14



So the sum of the shear flows due to the direct load through the shear centre plus the shear flows due to the torque is the same as the shear flows due to the load acting through the box centre illustrated in Figure 15.

$$\text{From torque } q = \frac{T}{2A} = \frac{720 \times 10^3}{2 \times 18 \times 10^4} = 2 \text{ N/mm}$$

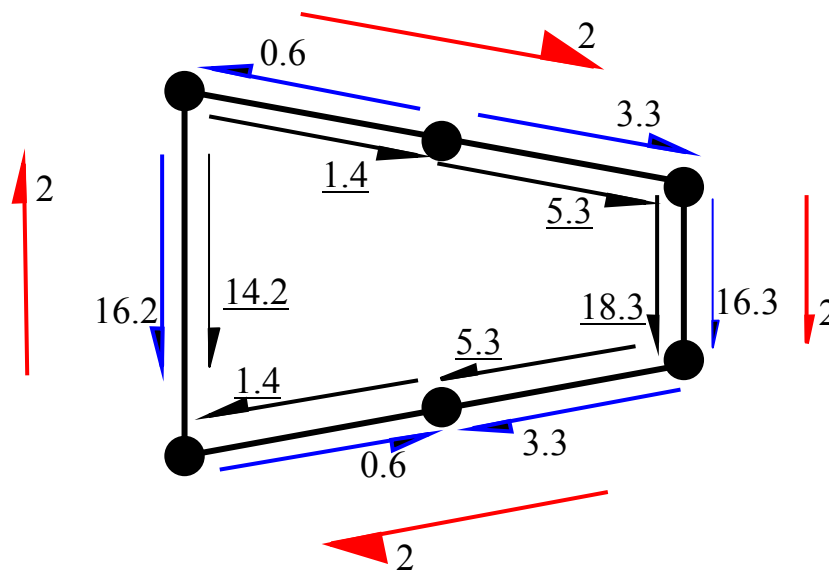


Figure 16

Figure 16 shows the shear flows due to the torque and load on the outside of the box, and the sum of the two, namely the net flow, is shown on the inside of the box. These are the same values as derived on page 13 – 9

See Tutorial Question 4

MULTICELL BOX SECTIONS

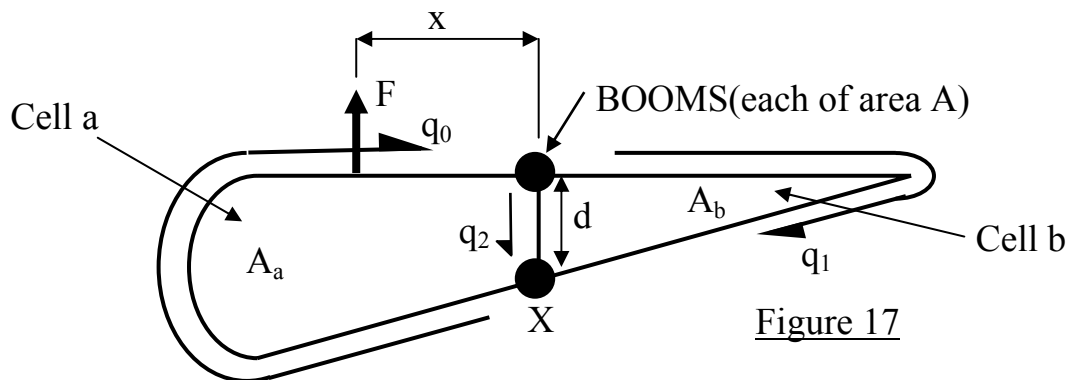


Figure 17

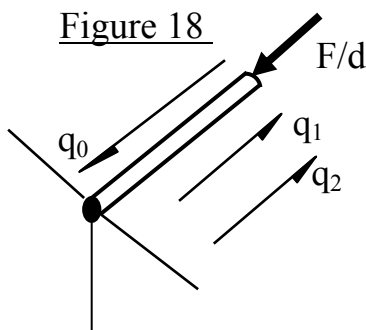
Figure 17 shows a simple two cell box loaded with force F as shown. The box is idealised with two booms of equal areas separated by a distance d . In order to determine the skin shear flows due to bending and twisting the following procedure can be used:-

- (a) Calculate the second moment of area of the section, $I = \sum Ah^2$
 $I = 2A \times (d/2)^2$

Now $\Delta P / \Delta L = F(A\bar{y}) / I$ thus $\Delta P / \Delta L = \frac{FA(d/2)}{2A(d/2)^2} = F/d$

Figure 18 shows the shear flows and end loads on the top boom,

Figure 18



for spanwise equilibrium $q_0 + F/d - q_1 - q_2 = 0 \dots (\text{equ.1})$

Taking moments about X : $Fx + 2A_a q_0 + 2A_b q_1 = 0 (\text{equ.2})$

Using the fact that the twist of each cell must be the same for compatibility, then a third equation can be obtained.

therefore $\theta_a = \theta_b$

$$\frac{L}{2A_a G_a} \sum \frac{q \Delta L}{t_a} = \frac{L}{2A_b G_b} \sum \frac{q \Delta L}{t_b}$$

assuming that in this case L and G are the same for each cell then:-

$$\frac{1}{A_a} \left(\frac{q_0 L_0}{t_0} + \frac{q_2 L_2}{t_2} \right) = \frac{1}{A_b} \left(\frac{q_1 L_1}{t_1} - \frac{q_2 L_2}{t_2} \right) \dots\dots\dots (\text{equ.3})$$

hence solving equations (1), (2) and (3) simultaneously the skin shear flows can be determined.

This solution method can be applied to boxes with any number of cells.

An iterative solution method can also be used to determine the shear flows in box sections which is more appropriate to multi-boom and multi-cell structures. An iterative method is also amenable to computerisation.

See tutorial question 8

EFFECT OF A CUT-OUT ON BEAM STRESSES

This is illustrated by means of an example.

The structural part of a wing consists of a 3 bay rectangular section box as shown in Figure 18.

The skin on the lower surface of the central bay has been removed and the box is subjected to a torque of 20 kNm at the free end.

Calculate the shear flows in the skin panels and spar webs, the loads in the corner flanges and the forces in the ribs on each side of the cut-out.

Assume that the flanges carry all the direct loads and the panels and webs only shear.

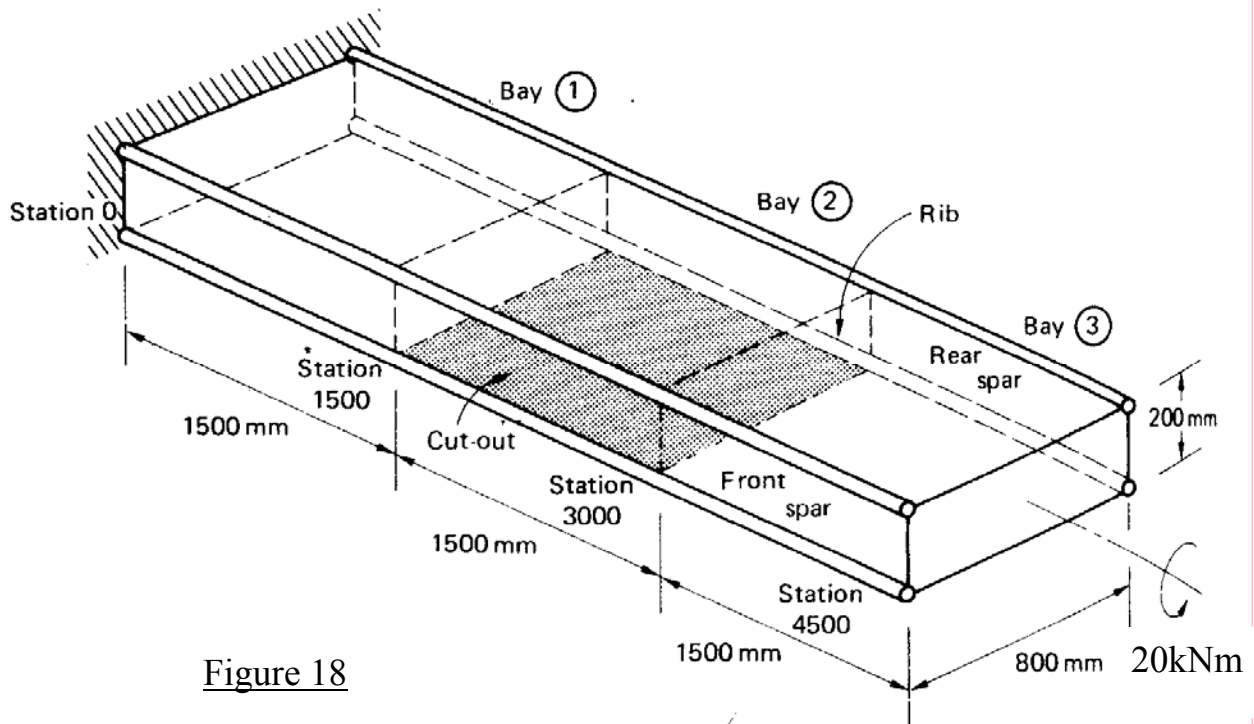


Figure 18

If the box was continuous then the flanges would be unloaded and the shear flows in the panels and webs would be :-

$$q = \frac{T}{2A} = \frac{20 \times 10^6}{2 \times 200 \times 800} = 62.6 \text{ N/mm} \dots \dots \dots \text{equation 1}$$

The removal of the lower skin panel in bay 2 results in a torsionally weak channel section for the length of bay 2 which must still transmit the applied torque to bay 1. As the front and rear spars are attached to torsionally stiff closed boxes then it will

be assumed that the torque is transmitted across the bay by differential bending of the spars. The bending moment in each spar is resisted by the flange loads P as shown in Figure 19(a).

Now the shear loads in the front and rear spars form a couple at any position in bay 2 which is equal to the applied torque. Thus from Figure 19(b):-

$$800S = 20 \times 10^6 \text{ N mm}$$

$$S = 25,000 \text{ N}$$

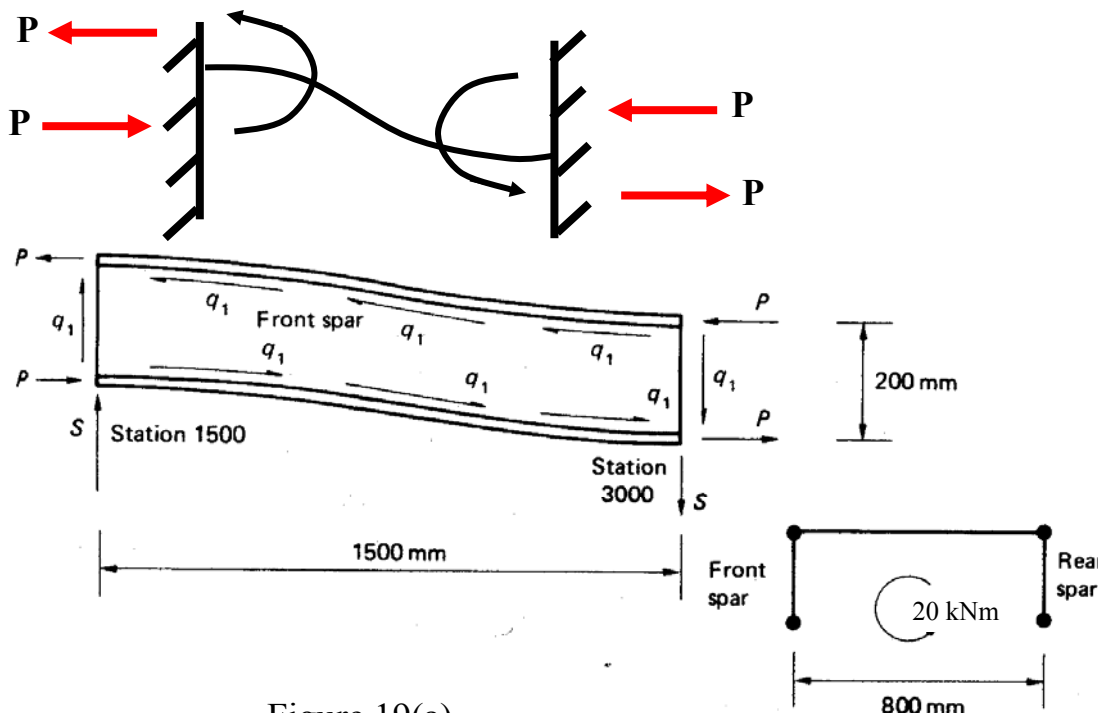


Figure 19(a)

Figure 19(b)

The shear flow q_1 , Figure 19(a), is given by:-

$$q_1 = 25,000/200 = 125 \text{ N/mm}$$

in the middle of bay 2 a point of contraflexure occurs where there is no bending moment. Thus:-

$$200 \times P = 25,000 \times 750 \text{ N mm} \quad P = 93,750 \text{ N}$$

Additionally P can be found by considering the equilibrium of either of the spar flanges, ie:-

$$2 \times P = 1500q_1 = 1500 \times 125 = 197,500 \text{ N} \quad \text{thus } P = 93,750 \text{ N}$$

The flange loads P are reacted by loads in the flanges of bays 1 and 3. These flange loads are transmitted to the adjacent webs and panels as shown in Figure 20 for bay 3 and thus modify the shear flow distribution given by equation 1.

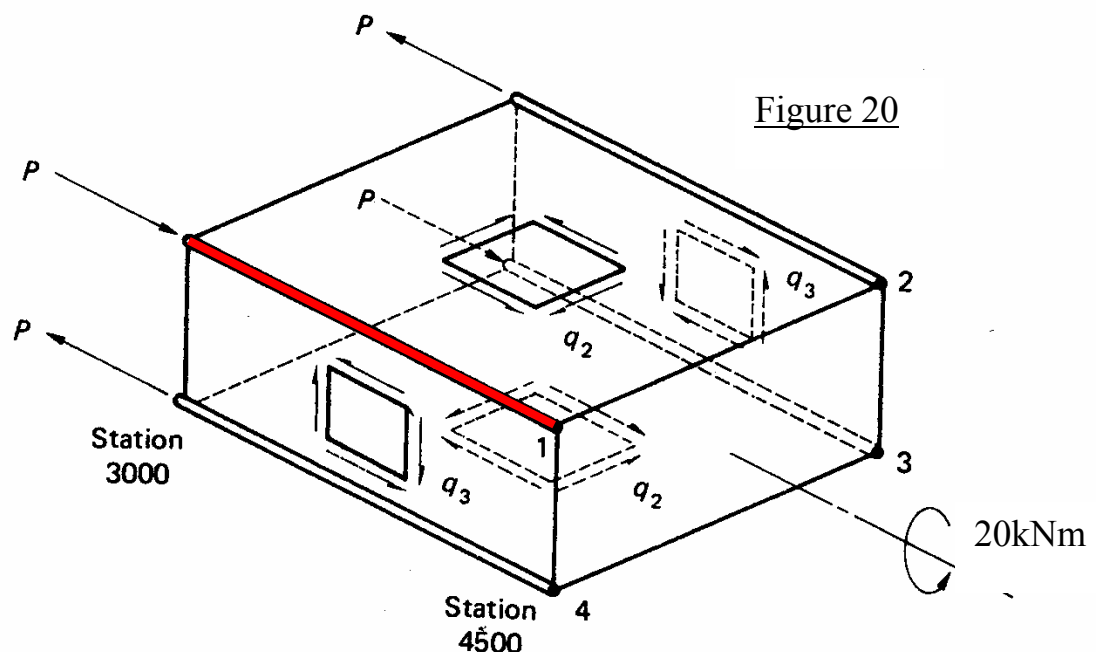


Figure 20

For equilibrium of flange 1:-

$$-1500q_2 + 1500q_3 = P = 93750 \text{ N}$$

$$-q_2 + q_3 = 62.6 \dots\dots\dots(2)$$

now the resultant of the shear flows q_2 and q_3 must be equivalent to the applied torque .hence, for moments about the lower front corner of bay 1 :-

$$200 \times 800q_2 + 800 \times 200q_3 = 20 \times 10^6 \text{ N mm}$$

$$q_2 + q_3 = 125 \dots\dots\dots(3)$$

solving equations (2) and(3) we get **$q_2 = 93.8 \text{ N/mm}$ and $q_3 = 31.2 \text{ N/mm}$**

Comparing these results with equation 1, the shear flows in the top and bottom panels are increased by 1.5 and decreased in the shear webs by a factor of 0.5.

See Tutorial Question 9

Tutorial

Question 1

Repeat example 1 but with the vertical sides being 4mm thick, the outer dimensions remaining the same.

Question 2

The profile shown in Figure Q2 forms part of a wing. The section is subjected to a shear force at its shear centre of 15.6 kN.

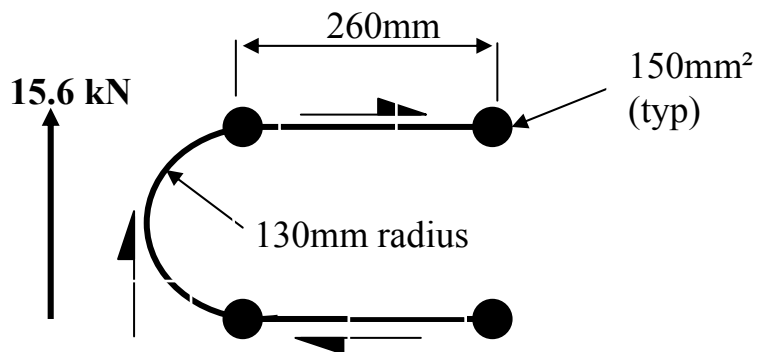


Figure Q2

- Given that the section has four idealised booms, determine the shear flows in the three panels.
- Determine the location of the shear centre, measured from the left hand booms and state its importance.

Question 3

A cantilevered, idealised, single cell box beam is shown in Figure Q3. A vertical 15kN load acts at point C. The sectional area of booms A and F is 900mm^2 , B and E is 150mm^2 , and C and D 600mm^2 . Using the shear flow notation shown determine:-

- the shear flows in the skins and the corresponding shear stresses
- the maximum bending stress in boom C if the beam length is 1.5m

Take any moments required about point F.

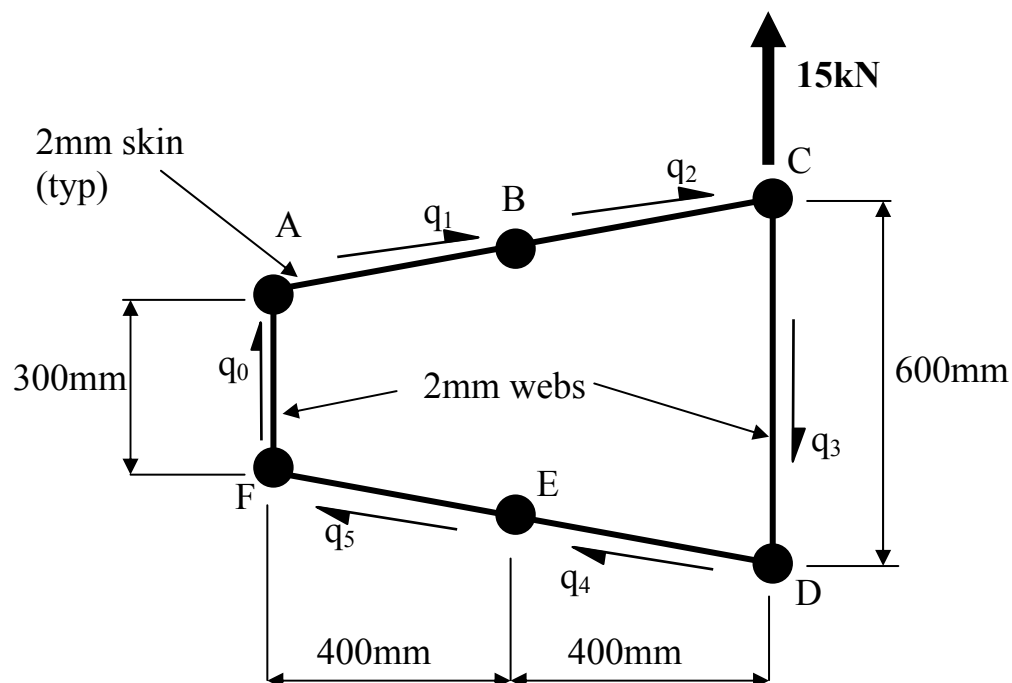


Figure Q3

Question 4

For the box section shown in Figure 9 calculate the value of the shear flow q_0 if a upward vertical load of 100 kN is applied at boom C.

Question 5

A glider wing section is shown in Figure Q5 consisting of a nose skin and vertical web joined together with two booms of equal area. The applied loads are a vertical shear force of 4.5kN and a pure torque of 1kNm.

Determine:-

- The shear flows in the skin and web due to these loads and the corresponding shear stresses
- The twist per metre length of wing section.
- Determine if the vertical web could buckle under the applied loads.

Take G for birch ply as 4kN/mm^2 and E as 9 kN/mm^2

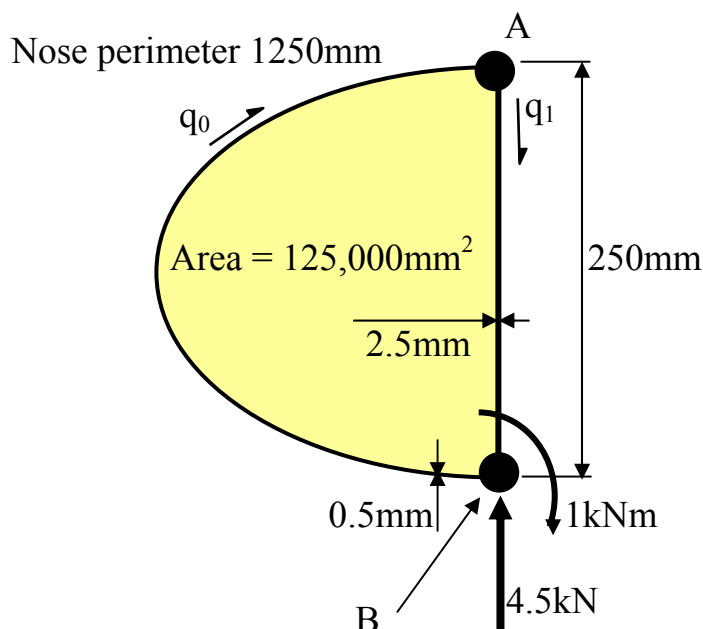


Figure Q5

ANS: (a) $q_0 = -4\text{N/mm}$, $q_1 = 14\text{N/mm}$
 (b) 0.49 degrees
 (c) $\tau_b = 4.9\text{N/mm}^2$

Question 6

An idealised single cell beam section is shown in Figure Q6. It is symmetrical about the X axis and is subjected to a shear load of 50 kN at a boom position C.

Calculate:-

- the shear flow in the booms
- the shear flow in the skins of the box (use the shear flow convention shown in Figure P2 and take any moments required about point C)
- the maximum skin shear stress
- the minimum skin shear stress reserve factor if the maximum allowable skin shear stress is 90 N/mm^2 and a factor of safety of 1.5 is required

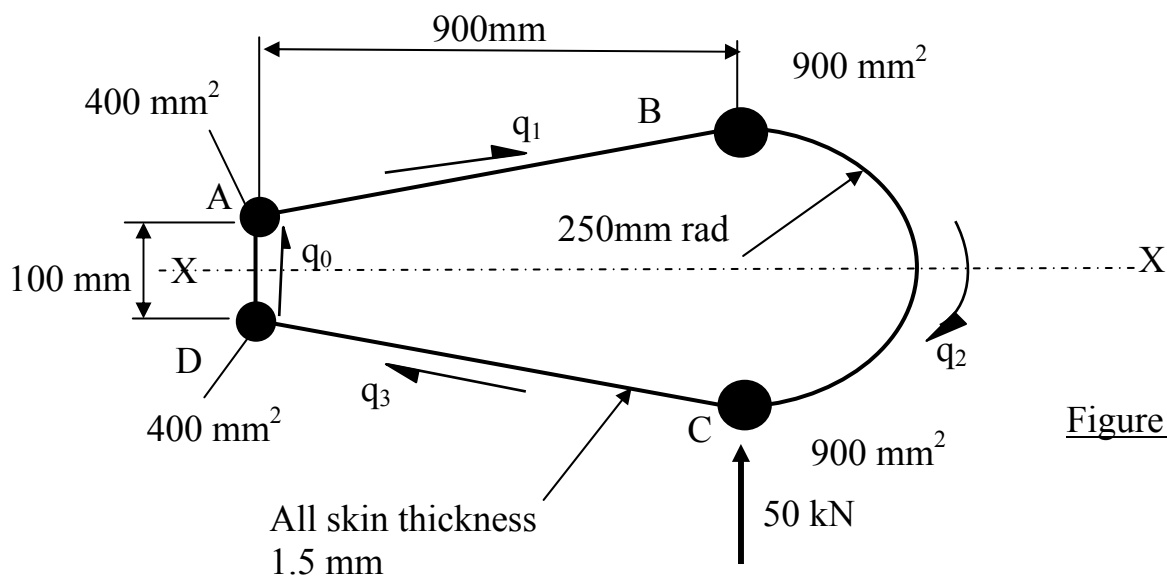


Figure Q6

ANS: (i) $\pm 8.7 \text{ N/mm}$, $\pm 98.3 \text{ N/mm}$
(ii) $q_0 = -33.9 \text{ N/mm}$, $q_1 = -25.1$, $q_2 = 73.1$
(iii) 48.8 N/mm^2
(iv) 1.23

Question 7

An idealised thin walled symmetrical box section is shown in Figure Q7
Assuming that the sheet material transmits shear only and the booms carry the end load; determine the sheet stress distribution for :-

- a pure torque of 1 kNm applied anticlockwise in the plane of the cross section.
- A vertical upward load of 1kN applied in line with the left hand web a-d.

Answer: (a) $q_0 = 14.15 \text{ N/mm}$ (b) $q_0 = 3.05 \text{ N/mm}$

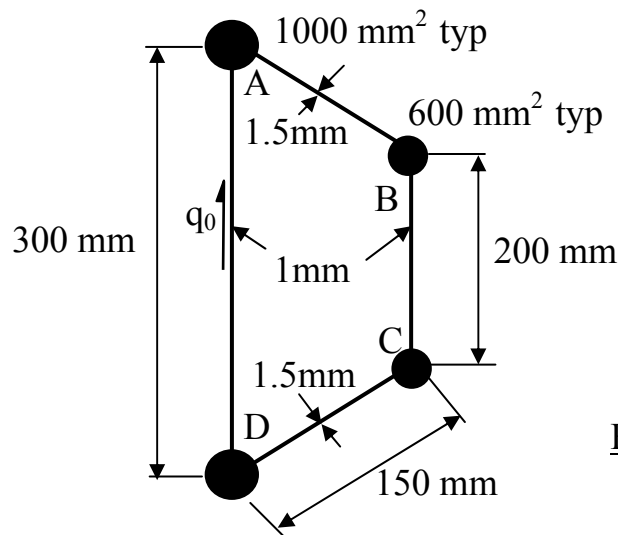


Figure Q7

Question 8

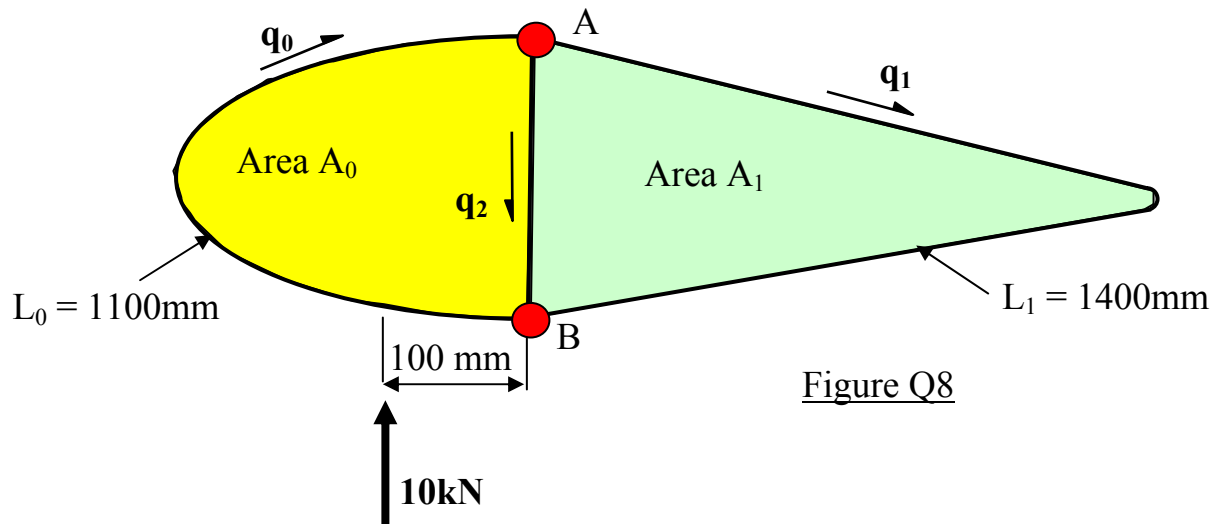


Figure Q8

Figure Q8 shows a two cell beam with two booms A and B of equal areas. Calculate the value of the shear flows q_0 , q_1 , and q_2 .

Web thickness = 2 mm, web depth = 200 mm

$A_0 = 1000 \text{ mm}^2$

$A_1 = 500 \text{ mm}^2$

Skin thickness = 1 mm

Question 9

A wing box is considered to be built-in at one end and carries a shear load of 2000 N through its shear centre and a torque of 1000 Nm as shown in Figure Q9(a). There is no skin panel in the upper surface of the inboard bay 1.

Figure 9(b) shows the shear flow conventions to be used

Calculate the shear flows q_1 , q_2 and q_3 .

(a) due to the applied torque only.

(b) due to the applied shear load only.

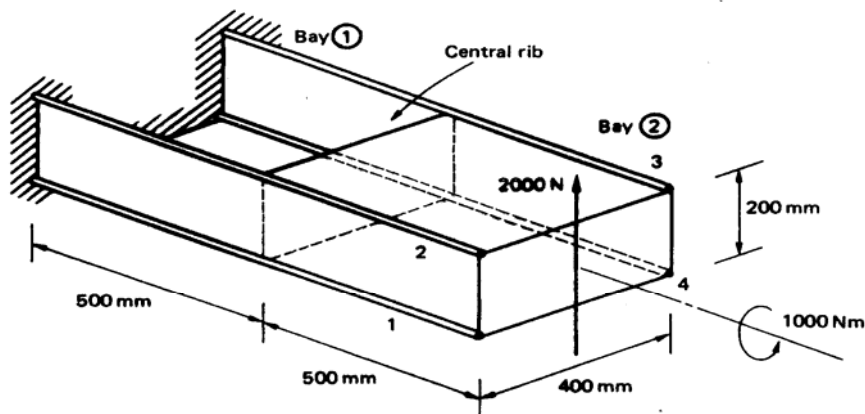


Figure Q9(a)

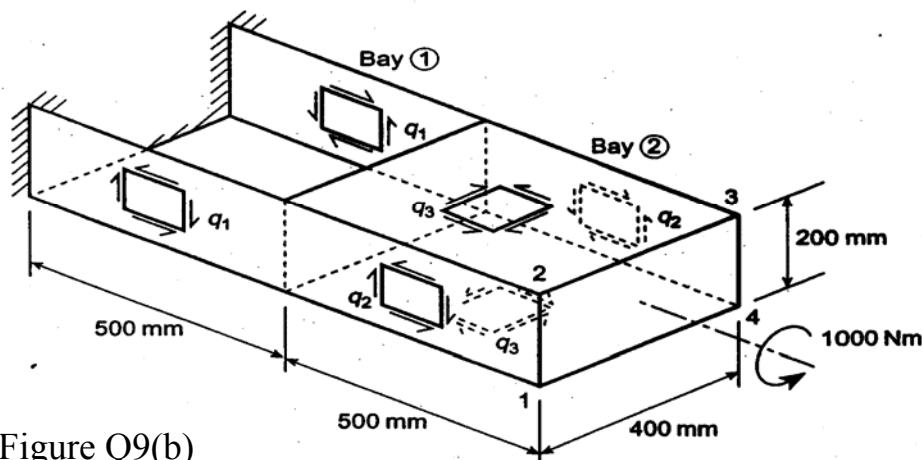


Figure Q9(b)